

MATH40004 - Calculus and Applications

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Autumn 2025 – Spring 2026
Compiled on May 13, 2026

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Be mindful of any mistakes or omissions if using these notes for self-study or
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Introduction

This course assumes mastery of A-Level Mathematics (and Further Mathematics), in particular differentiation, integration and associated techniques, curve sketching, and elementary mechanics. Ultimately, this is a methods-based course, and while a certain level of mathematical communication and clarity is expected, we do not focus on rigorously proving facts about calculus in this course (that is the purpose of MATH40002, Analysis I). Any proofs that are omitted will be covered in MATH40002 (most of the time, in this course, we are just interested in the result and the consequences).

Chapter 1

Limits and Continuity

1.1 Limits of Functions

The fundamental concept of calculus is the limit. We formalise the notion of a function approaching a specific value.

Definition 1.1 (ε - δ definition of a limit). Let f be a function defined on an open interval containing x_0 , except possibly at x_0 itself. We say that $\lim_{x \rightarrow x_0} f(x) = \ell$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x ,

$$0 < |x - x_0| < \delta \implies |f(x) - \ell| < \varepsilon.$$

Example 1.2. We prove that $\lim_{x \rightarrow 2} (3x - 1) = 5$. Given $\varepsilon > 0$, we want to find $\delta > 0$ such that $0 < |x - 2| < \delta$ implies $|(3x - 1) - 5| < \varepsilon$. We manipulate the consequent:

$$|(3x - 1) - 5| = |3x - 6| = 3|x - 2|.$$

To ensure this is strictly less than ε , we require $3|x - 2| < \varepsilon$, which is equivalent to $|x - 2| < \frac{\varepsilon}{3}$. Therefore, we choose $\delta = \frac{\varepsilon}{3}$. Then for all x satisfying $0 < |x - 2| < \delta$, we have $|(3x - 1) - 5| = 3|x - 2| < 3\delta = \varepsilon$.

Theorem 1.3 (Algebraic properties of limits). Suppose $\lim_{x \rightarrow x_0} f(x) = L$ and $\lim_{x \rightarrow x_0} g(x) = M$. Then:

1. *Sum:* $\lim_{x \rightarrow x_0} (f(x) + g(x)) = L + M$
2. *Product:* $\lim_{x \rightarrow x_0} (f(x)g(x)) = LM$
3. *Quotient:* $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{L}{M}$, provided $M \neq 0$
4. *Scalar multiple:* $\lim_{x \rightarrow x_0} (cf(x)) = cL$ for any constant $c \in \mathbb{R}$

Definition 1.4 (Limit at infinity). We say that $\lim_{x \rightarrow \infty} f(x) = \ell$ if for every $\varepsilon > 0$, there exists an $A > 0$ such that for all x ,

$$x > A \implies |f(x) - \ell| < \varepsilon.$$

Example 1.5. We prove $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$. Given $\varepsilon > 0$, we require $|\frac{1}{x} - 0| < \varepsilon$, which means $\frac{1}{x} < \varepsilon$ for $x > 0$. Rearranging gives $x > \frac{1}{\varepsilon}$. We choose $A = \frac{1}{\varepsilon}$. For all $x > A$, we have $x > \frac{1}{\varepsilon}$, so $\frac{1}{x} < \varepsilon$.

Definition 1.6. We say that $\lim_{x \rightarrow x_0} f(x) = \infty$ if for every $B > 0$, there exists a $\delta > 0$ such that for all x ,

$$0 < |x - x_0| < \delta \implies f(x) > B.$$

Definition 1.7 (One-sided limits). The right-sided limit $\lim_{x \rightarrow x_0^+} f(x) = \ell$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x ,

$$0 < x - x_0 < \delta \implies |f(x) - \ell| < \varepsilon.$$

The left-sided limit $\lim_{x \rightarrow x_0^-} f(x) = \ell$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x ,

$$0 < x_0 - x < \delta \implies |f(x) - \ell| < \varepsilon.$$

A two-sided limit exists if and only if both one-sided limits exist and are equal.

Theorem 1.8 (Squeezing property). Suppose $g(x) \leq f(x) \leq h(x)$ for all x in an open interval containing x_0 , except possibly at x_0 itself. If $\lim_{x \rightarrow x_0} g(x) = \lim_{x \rightarrow x_0} h(x) = \ell$, then $\lim_{x \rightarrow x_0} f(x) = \ell$.

Example 1.9. Evaluate $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$. Since $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$ for all $x \neq 0$, we multiply the inequality by $|x|$ to obtain:

$$-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|.$$

We know $\lim_{x \rightarrow 0} (-|x|) = 0$ and $\lim_{x \rightarrow 0} |x| = 0$. By the comparison test, $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$.

1.2 Continuity

Definition 1.10 (Continuity at a point). A function f is continuous at x_0 if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all x in the domain of f ,

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

Equivalently, this requires that $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

Remark. For f to be continuous at x_0 , three conditions must be satisfied: $f(x_0)$ must be defined, $\lim_{x \rightarrow x_0} f(x)$ must exist, and these two values must be equal.

Example 1.11. Let $f(x)$ be defined as:

$$f(x) = \begin{cases} \frac{x^2-1}{x-1} & x \neq 1 \\ 2 & x = 1 \end{cases}$$

We check for continuity at $x = 1$. We have $f(1) = 2$. Taking the limit as $x \rightarrow 1$, we note that $x \neq 1$, so we can divide by $x - 1$:

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 2.$$

Since $\lim_{x \rightarrow 1} f(x) = f(1)$, the function is continuous at $x = 1$.